

Statics Internal Forces: Intro to Shear and Bending Moment Diagrams (V and M Diagrams)

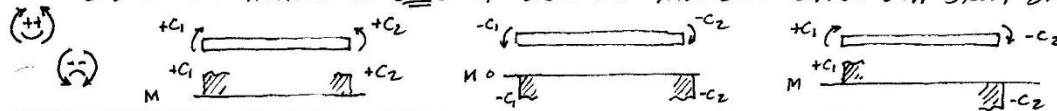
- ① POINT LOADS AND SUPPORTS PRODUCE JUMPS IN THE SHEAR DIAGRAM
IN THE DIRECTION OF THE POINT LOAD (OR SUPPORT)

RULES FOR V & M DIAGRAM

- ② COUPLES CORRESPOND TO JUMPS IN THE MOMENT DIAGRAM

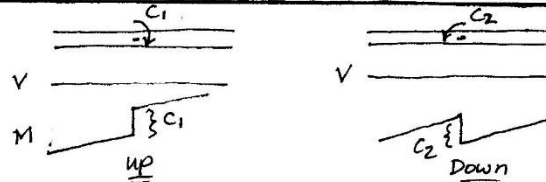
Couples acting at ends of beams are treated differently than those in interior.

- (a) COUPLES ACTING AT ENDS OF BEAMS: Moment curve will start or end at these values.



- (b) COUPLES ACTING IN INTERIOR OF BEAM

CW $\curvearrowright$ COUPLE PRODUCES (+) JUMP
CCW $\curvearrowleft$ " " (-) JUMP



NOTICE:
No change in V curve.

- ③ SLOPE AND AREA RELATIONS

$$\Delta V = -\int_A^B w dx$$

ΔV over interval AB =
Area under load (w) curve

$$\frac{dV}{dx} = -W$$

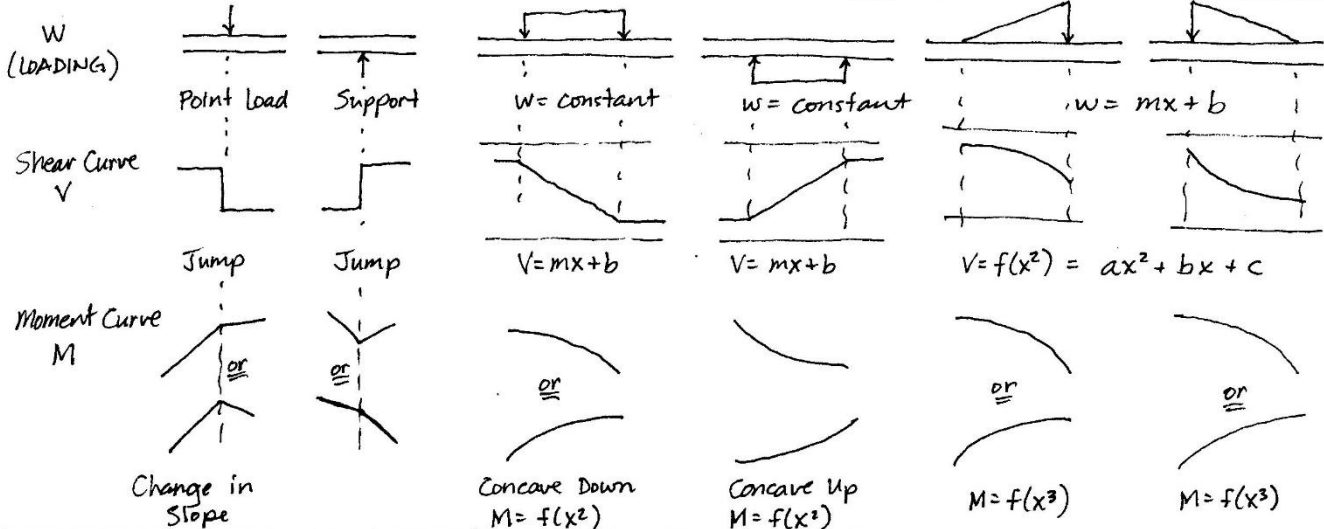
Slope of V
curve at a point
= -W at that point

$$\Delta M = \int_A^B V dx$$

ΔM over
interval AB
= Area under V curve

$$\frac{dM}{dx} = V$$

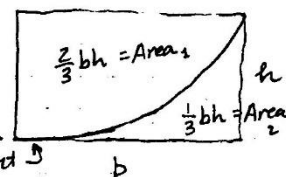
Slope of M curve
at a point
= V at that point



MATTERS OF STYLE

- Find areas of each distributed load on beam, and areas of each interval on V curve. Write these on area and enclose w/ circle, e.g. (10), (15) etc.
- Label all maxima, minima, & significant points.
- Shade areas.
- Label graphs w/ V & M and give units.

A useful formula for a parabolic area:



WRITING V & M EQNS

Must pass a section and write $\sum F_y$ and $\sum M$ equations between each place where a support, point load, distributed load, or couple occurs, and in each section where a distributed load acts. At right, would need 7 FBD's, and $\sum F_y \rightarrow V(x)$ and $\sum M \rightarrow M(x)$ at each section.

